

WASTE PROCESSING TECHNIQUES & SOURCE REDUCTION, PRODUCT RECOVERY & RECYCLING:

Purpose of processing, mechanical volume and size reduction, component separation, drying and dewatering. Source Reduction, Product Recovery and Recycling: basics, purpose, implementation monitoring and evaluation of source reduction, significance of recycling, planning of a recycling programme, recycling programme elements, commonly recycled materials and processes, a case study.

MODULE – 4

PURPOSE OF PROCESSING:

The purposes of processing, are:

- i. **Improving efficiency of SWM system:** Various processing techniques are available to improve the efficiency of SWM system. For example, before waste papers are reused, they are usually baled to reduce transporting and storage volume requirements. In some cases, wastes are baled to reduce the haul costs at disposal site, where solid wastes are compacted to use the available land effectively. If solid wastes are to be transported hydraulically and pneumatically, some form of shredding is also required. Shredding is also used to improve the efficiency of the disposal site.
- ii. **Recovering material for reuse:** Usually, materials having a market, when present in wastes in sufficient quantity to justify their separation, are most amenable to recovery and recycling. Materials that can be recovered from solid wastes include paper, cardboard, plastic, glass, ferrous metal, aluminium and other residual metals.
- iii. **Recovering conversion products and energy:** Combustible organic materials can be converted to intermediate products and ultimately to usable energy. This can be done either through incineration, pyrolysis, composting or bio-digestion. Initially, the combustible organic matter is separated from the other solid waste components. Once separated, further processing like shredding and drying is necessary before the waste material can be used for power generation.

MECHANICAL VOLUME AND SIZE REDUCTION:

Mechanical volume and size reduction is an important factor in the development and operation of any SWM system. The main purpose is to reduce the volume (amount) and size of waste, as compared to its original form, and produce waste of uniform size.

Volume reduction or compaction:

Volume reduction or compaction refers to densifying wastes in order to reduce their volume. Some of the benefits of compaction include:

- reduction in the quantity of materials to be handled at the disposal site;
- improved efficiency of collection and disposal of wastes;
- increased life of landfills;

- Economically viable waste management system.

However, note the following disadvantages associated with compaction:

- poor quality of recyclable materials sorted out of compaction vehicle;
- difficulty in segregation or sorting (since the various recyclable materials are mixed and compressed in lumps);
- Bio-degradable materials (e.g., leftover food, fruits and vegetables) destroy the value of paper and plastic material.

Equipment used for compaction Based on their mobility, we can categorise the compaction equipment used in volume reduction under either of the following:

Stationary equipment: This represents the equipment in which wastes are brought to, and loaded into, either manually or mechanically. In fact, the compaction mechanism used to compress waste in a collection vehicle, is a stationary compactor. According to their application, stationary compactors can be described as light duty (e.g., those used for residential areas), commercial or light industrial, heavy industrial and transfer station compactors. Usually, large stationary compactors are necessary, when wastes are to be compressed into:

- steel containers that can be subsequently moved manually or mechanically;
- chambers where the compressed blocks are banded or tied by some means before being removed;
- chambers where they are compressed into a block and then released and hauled away untied;
- transport vehicles directly.

Movable equipment: This represents the wheeled and tracked equipment used to place and compact solid wastes, as in a sanitary landfill.

Types of Compaction Equipment

Location or Operation	Type of Compactor Stationary/residential	Remarks
Solid waste generation points	Vertical	Vertical compaction ram may be used; may be mechanically or hydraulically operated, usually hand-fed; wastes compacted into corrugated box containers, or paper or plastic bags; used in medium and high-rise apartments.
	Rotary	Ram mechanism used to compact waste into paper or plastic bags on rotating platform, platform rotates as containers are filled; used in medium and high-rise apartments.
	Bag or extruder	Compactor can be chute fed; either vertical or horizontal rams; single or continuous multi-bags; single bag must be replaced and continuous bags must be tied off and replaced; used in medium and high-rise apartments.

Location or Operation	Type of Compactor Stationary/residential	Remarks
	Under counter	Small compactors used in individual residences and apartment units; wastes compacted into special paper bags; after wastes are dropped through a panel door into a bag and door is closed, they are sprayed for odour control; button is pushed to activate compaction mechanism.
	Stationary/commercial	Compactor with vertical and horizontal ram; wastes compressed into steel containers; compressed wastes are manually tied and removed; used in low, medium and high-rise apartments, commercial and industrial facilities.
Collection	Stationary/packers	Collection vehicles equipped with compaction mechanism.
Transfer and/or processing station	Stationary/transfer trailer	Transfer trailer, usually enclosed, equipped with self-contained compaction mechanism.
	Stationary low pressure	Wastes are compacted into large containers.
	Stationary high pressure	Wastes are compacted into dense bales or other forms.
Disposal site	Movable wheeled or tracted equipment	Specially designed equipment to achieve maximum compaction of wastes.
	Stationary/track mounted	High-pressure movable stationary compactors used for volume reduction at a disposal site.

Compactors According to their compaction pressure, we can divide the compactors used at transfer stations as follows:

- (i) **Low-pressure (less than 7kg/cm²) compaction:** This includes those used at apartments and commercial establishments, bailing equipment used for waste papers and cardboards and stationary compactors used at transfer stations. In low-pressure compaction, wastes are compacted in large containers. Note that portable stationary compactors are being used increasingly by a number of industries in conjunction with material recovery options, especially for waste paper and cardboard.
- (ii) **High-pressure (more than 7kg/cm²) compaction:** Compact systems with a capacity up to 351.5 kg/cm² or 5000 lb/in² come under this category. In such systems, specialised compaction equipment are used to compress solid wastes into blocks or bales of various sizes. In some cases, pulverised wastes are extruded after compaction in the form of logs. The volume reduction achieved with these high-pressure compaction systems varies with the characteristics of the waste.

$$\text{Volume Reduction (\%)} = \frac{V_i - V_f}{V_i} \times 100$$

$$\text{Compaction ratio} = \frac{V_i}{V_f}$$

where V_i = volume of waste before compaction, m³ and V_f = volume of waste after compaction, m³

The relationship between the compaction ratio and percent of volume reduction is important in making a trade-off analysis between compaction ratio and cost. Other factors that must be considered are final density of waste after compaction and moisture content. The moisture content that varies with location is another variable that has a

major effect on the degree of compaction achieved. In some stationary compactors, provision is made to add moisture, usually in the form of water, during the compaction process.

Selection of compaction equipment:

To ensure effective processing, we need to consider the following factors, while selecting compaction equipment:

- Characteristics such as size, composition, moisture content, and bulk density of the waste to be compacted. Method of transferring and feeding wastes to the compactor, and handling.
- Potential uses of compacted waste materials. Design characteristics such as the size of loading chamber, compaction pressure, compaction ratio, etc.
- Operational characteristics such as energy requirements, routine and specialised maintenance requirement, simplicity of operation, reliability, noise output, and air and water pollution control requirement.
- Site consideration, including space and height, access, noise and related environmental limitations.

Size reduction or shredding

This is required to convert large sized wastes (as they are collected) into smaller pieces. Size reduction helps in obtaining the final product in a reasonably uniform and considerably reduced size in comparison to the original form. But note that size reduction does not necessarily imply volume reduction, and this must be factored into the design and operation of SWM systems as well as in the recovery of materials for reuse and conversion to energy. In the overall process of waste treatment and disposal, size reduction is implemented ahead of:

- land filling to provide a more homogeneous product. This may require less cover material and less frequent covering than that without shredding. This can be of economic importance, where cover material is scarce or needs to be brought to the landfill site from some distance.
- recovering materials from the waste stream for recycling.
- baling the wastes – a process sometimes used ahead of long distance transport of solid wastes – to achieve a greater density.
- making the waste a better fuel for incineration waste energy recovery facilities. (The size reduction techniques, coupled with separation techniques such as screening, result in a more homogeneous mixture of relatively uniform size, moisture content and heating value, and thereby improving the steps of incineration and energy recovery.
- reducing moisture, i.e., drying and dewatering of wastes

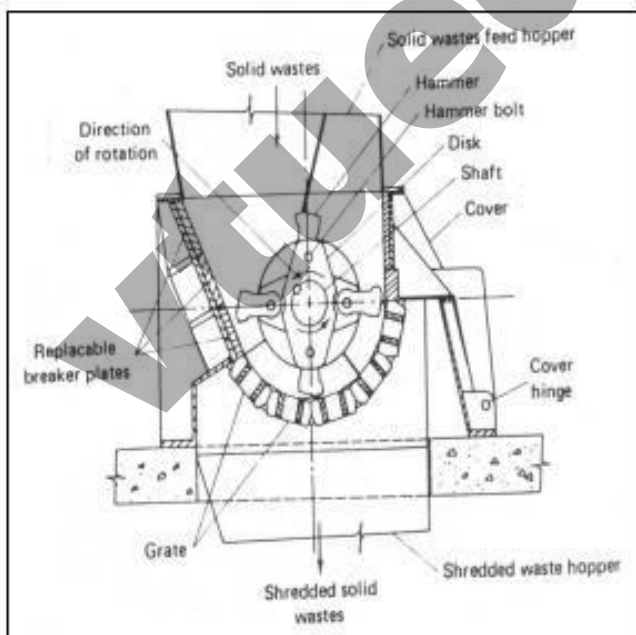
Size Reduction Equipment

Type	Mode of action	Application
Small grinders	Grinding, mashing	Organic residential solid wastes
Chippers	Cutting, slicing	Paper, cardboard, tree trimmings, yard waste, wood, plastics
Large grinders	Grinding, mashing	Brittle and friable materials, used mostly in industrial operation
Jaw crushers	Crushing, breaking	Large solids
Rasp mills	Shredding, tearing	Moistened solid wastes
Shredders	Shearing, tearing	All types of municipal wastes
Cutters, Clippers	Shearing, tearing	All types of municipal wastes
Hammer mills	Breaking, tearing, cutting, crushing	All types of municipal wastes, most commonly used equipment for reducing size and homogenizing composition of wastes
Hydropulper	Shearing, tearing	Ideally suited for use with pulpable wastes, including paper, wood chips. Used primarily in the papermaking industry. Also used to destroy paper records

The most frequently used shredding equipment are the following:

(i) Hammer mill: These are used most often in large commercial operations for reducing the size of wastes. Hammer mill is an impact device consisting of a number of hammers, fastened flexibly to an inner disk, which rotates at a very high speed:

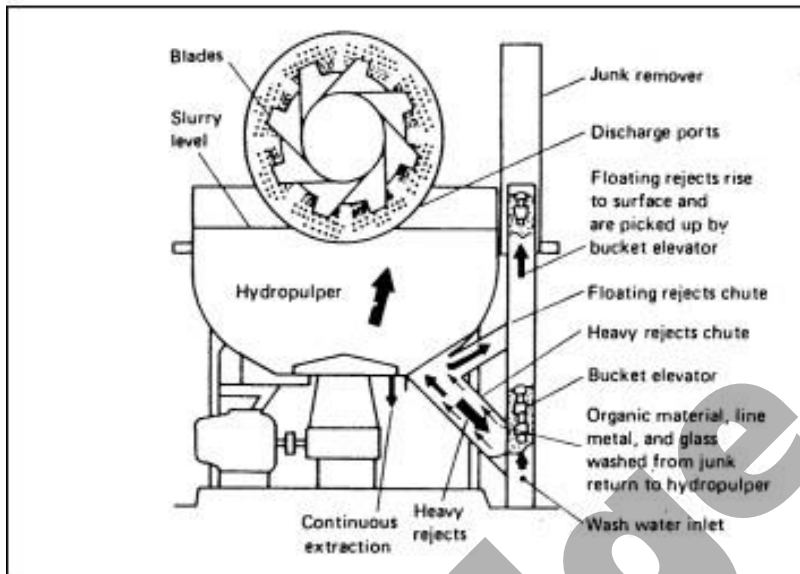
Hammer Mill: An Illustration



Solid wastes, as they enter the mill, are hit by sufficient force, which crush or tear them with a velocity so that they do not adhere to the hammers. Wastes are further reduced in size by being struck between breaker plates and/or cutting bars fixed around the periphery of the inner chamber. This process of cutting and striking action continues, until the required size of material is achieved and after that it falls out of the bottom of the mill.

ii. **Hydropulper:** An alternative method of size reduction involves the use of a hydropulper.

Hydropulper: An Illustration



Solid wastes and recycled water are added to the hydropulper. The highspeed cutting blades, mounted on a rotor in the bottom of the unit, convert pulpable and friable materials into slurry with a solid content varying from 2.5 to 3.5%. Metal, tins, cans and other non-pulpable or non-friable materials are rejected from the side of the hydropulper tank. The rejected material passes down a chute that is connected to a bucket elevator, while the solid slurry passes out through the bottom of the pulper tank and is pumped to the next processing operation.

Selection of size reduction equipment

- The factors that decide the selection of size reduction equipment include the following:
- The properties of materials before and after shredding.
- Size requirements for shredded material by component.
- Method of feeding shredders, provision of adequate shredder hood capacity (to avoid bridging) and clearance requirement between feed and transfer conveyors and shredders.
- Types of operation (continuous or intermittent).
- Operational characteristics including energy requirements, routine and specialised maintenance requirement, simplicity of operation, reliability, noise output, and air and water pollution control requirements.

- Site considerations, including space and height, access, noise and environmental limitations.
- Metal storage after size reduction for the next operation.

Chemical volume reduction

Chemical volume reduction is a method, wherein volume reduction occurs through chemical changes brought within the waste either through an addition of chemicals or changes in temperature. Incineration is the most common method used to reduce the volume of waste chemically, and is used both for volume reduction and power production. These other chemical methods used to reduce volume of waste chemically include *pyrolysis*, *hydrolysis* and chemical conversions.

Note that prior to size or volume reduction, component separation is necessary to avoid the problem of segregating or sorting recyclable materials from the mixed and compressed lumps of wastes and the poor quality of recyclable materials sorted out of compaction vehicles.

COMPONENT SEPARATION

Component separation is a necessary operation in which the waste components are identified and sorted either manually or mechanically to aid further processing. This is required for the:

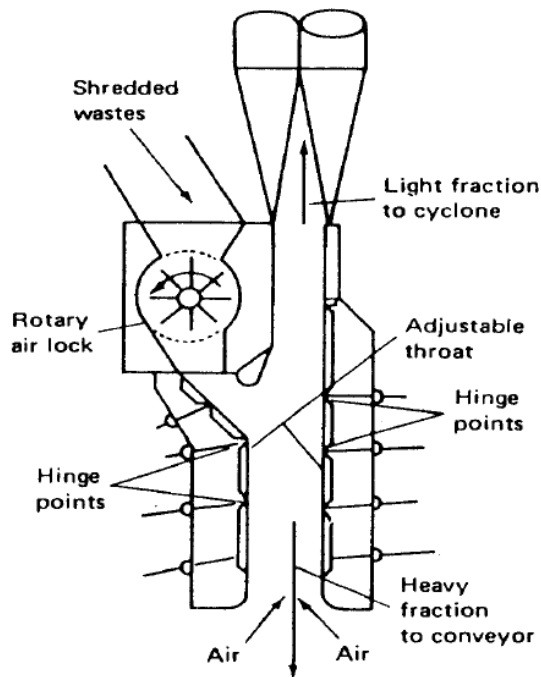
- recovery of valuable materials for recycling;
- preparation of solid wastes by removing certain components prior to incineration, energy recovery, composting and biogas production.

The most effective way of separation is manual sorting in households prior to collection. In many cities (e.g., Bangalore, Chennai, etc., in India), such systems are now routinely used. The municipality generally provides separate, easily identifiable containers into which the householder deposits segregated recyclable materials such as paper, glass, metals, etc. Usually, separate collections are carried out for the recyclable material. At curbside, separate areas are set aside for each of the recyclable materials for householders to deliver material – when there is no municipal collection system. In case the separation is not done prior to collection, it could be sorted out through mechanical techniques such as air separation, magnetic separation, etc., to recover the wastes.

Air separation

This technique has been in use for a number of years in industrial operations for segregating various components from dry mixture. Air separation is primarily used to separate lighter materials (usually organic) from heavier (usually inorganic) ones. The lighter material may include plastics, paper and paper products and other organic materials. Generally, there is also a need to separate the light fraction of organic material from the conveying air streams, which is usually done in a cyclone separator. In this technique, the heavy fraction is removed from the air classifier (i.e., equipment used for air separation) to the recycling stage or to land disposal, as appropriate. The light fraction may be used, with or without further size reduction, as fuel for incinerators or as compost material. There are various types of air classifiers commonly used, some of which are listed below:

Conventional chute type:

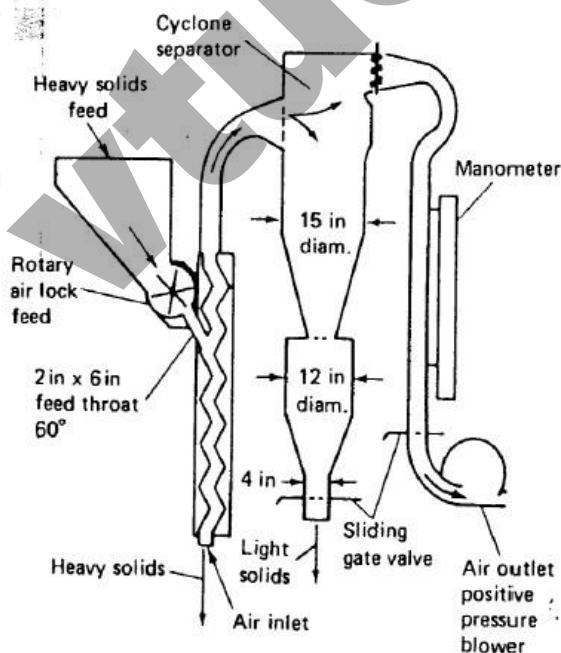


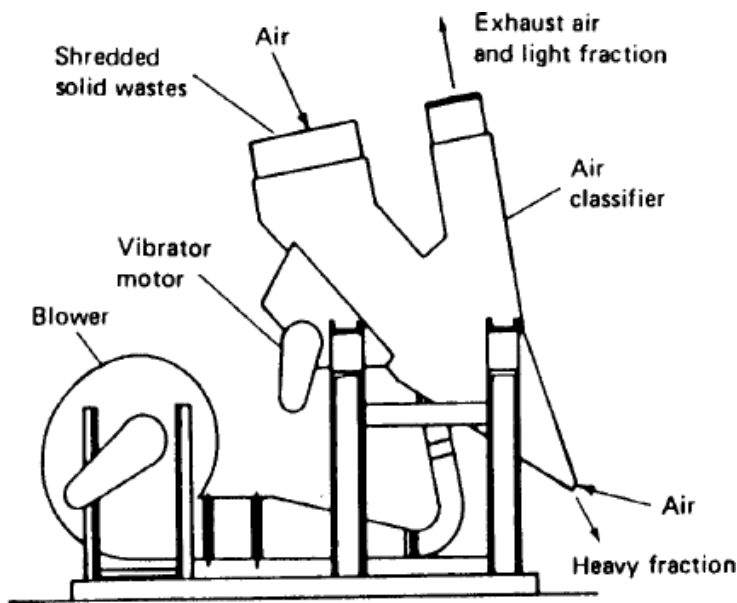
In this type, when the processed solid wastes are dropped into the vertical chute, the lighter material is carried by the airflow to the top while the heavier materials fall to the bottom of the chute. The control of the percentage split between the light and heavy fraction is accomplished by varying the waste loading rate, airflow rate and the cross section of chute. A rotary air lock feed mechanism is required to introduce the shredded wastes into the classifier.

Zigzag air classifier:

An experimental zigzag air classifier, shown in Figure below, consists of a continuous vertical column with internal zigzag deflectors through which air is drawn at a high rate:

Shredded wastes are introduced at the top of the column at a controlled rate, and air is introduced at the bottom of the column. As the wastes drop into the air stream, the lighter fraction is fluidised and moves upward and out of column, while the heavy fraction falls to the bottom. Best separation can be achieved through proper design of the separation chamber, airflow rate and influent feed rate.



Open inlet vibrator type:

In this type of air classifier, the separation is accomplished by a combination of the following actions:

- **Vibration:** This helps to stratify the material fed to the separator into heavy and light components. Due to this agitation, the heavier particles tend to settle at the bottom as the shredded waste is conveyed down the length of the separator.
- **Inertial force:** In this action, the air pulled in through the feed inlet imparts an initial acceleration to the lighter particle, while the wastes travel down the separator as they are being agitated.
- **Air pressure:** This action refers to the injection of fluidising air in two or more high velocity and low mass flow curtains across the bed. A final stripping of light particles is accomplished at the point where the heavy fraction discharges from the elutriators. It has been reported that the resulting separation is less sensitive to particle size than a conventional vertical air classifier, be it of straight or zigzag design. An advantage of this classifier is that an air lock feed mechanism is not required and wastes are fed by gravity directly into the separator inlet.

Selection of air separation equipment

The factors that are to be considered for selecting air separation equipment include the following:

- Characteristics of the material produced by shredding equipment including particle size, shape, moisture content and fibre content.
- Material specification for light fraction.
- Methods of transferring wastes from the shredders to the air separation units and feeding wastes into the air separator.
- Characteristics of separator design including solids-to-air ratio, fluidising velocities, unit capacity, total airflow and pressure drop.
- Operational characteristics including energy requirement, maintenance requirement, simplicity of operation, proved performance and reliability, noise output, and air and water pollution control requirements.
- Site considerations including space and height access, noise and environmental limitations.

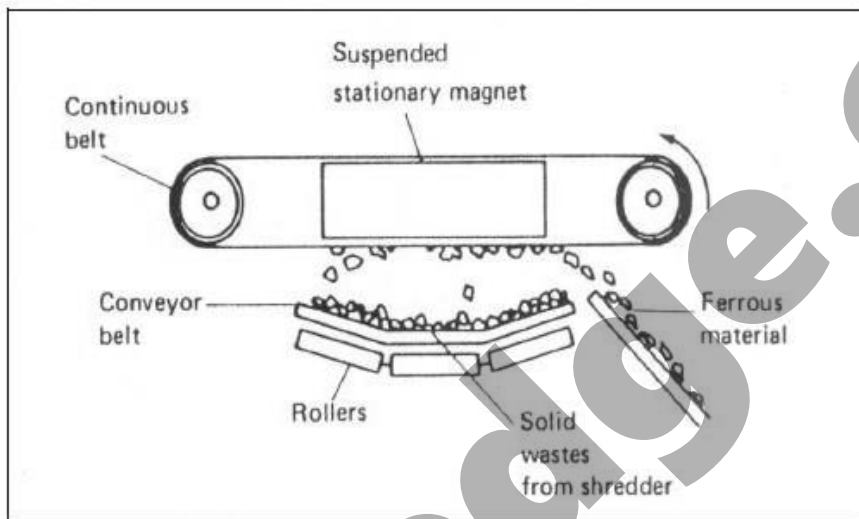
Magnetic separation:

The most common method of recovering ferrous scrap from shredded solid wastes involves the use of magnetic recovery systems. Ferrous materials are usually recovered either after shredding or before air classification. When wastes are mass-fired in incinerators, the magnetic separator is used to remove the ferrous material from the incinerator residue. Magnetic recovery systems have also been used at landfill disposal sites. The specific locations, where ferrous materials are recovered will depend on the objectives to be achieved, such as reduction of wear and tear on processing and separation equipment, degree of product purity achieved and the required recovery efficiency.

Equipment used for magnetic separation:

(i) **Suspended magnet:** In this type of separator, a permanent magnet is used to attract the ferrous metal from the waste stream. When the attracted metal reaches the area, where there is no magnetism, it falls away freely. This ferrous metal is then collected in a container. Figure 5.6 shows a typical suspended magnet:

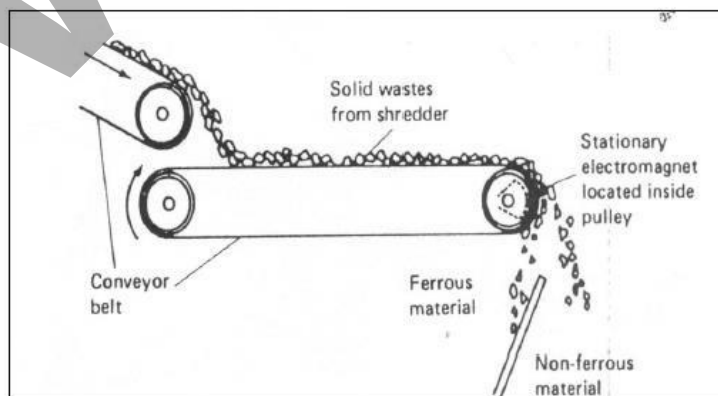
Suspended Type Permanent Magnetic Separator



This type of separation device is suitable for processing raw refuse, where separators can remove large pieces of ferrous metal easily from the waste stream.

(ii) **Magnetic pulley:** This consists of a drum type device containing permanent magnets or electromagnets over which a conveyor or a similar transfer mechanism carries the waste stream. The conveyor belt conforms to the rounded shape of the magnetic drum and the magnetic force pulls the ferrous material away from the falling stream of solid waste. It illustrates this type of magnetic separator:

Pulley Type Permanent Magnetic Separator



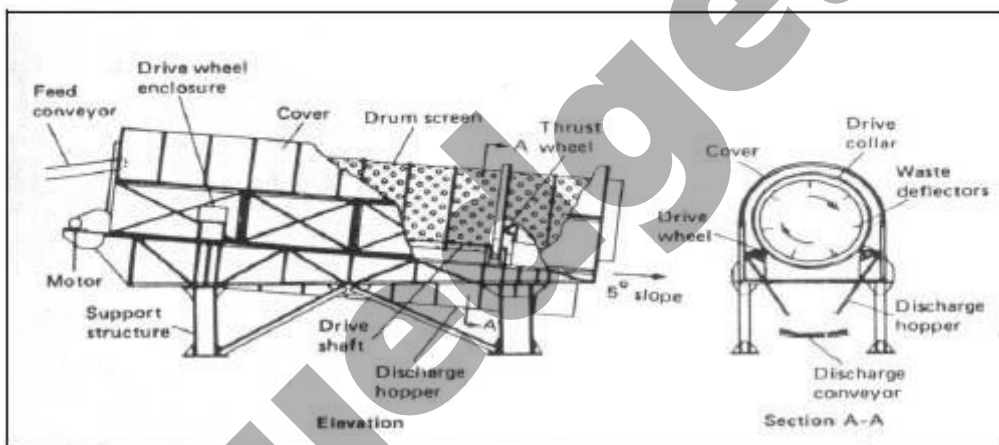
Selection of magnetic separation equipment

We must consider the following factors in the selection of magnetic separation equipment:

- Characteristics of waste from which ferrous materials are to be separated (i.e., the amount of ferrous material, the tendency of the wastes to stick to each other, size, moisture content, etc.)
- Equipment used for feeding wastes to separator and removing the separated waste streams. Characteristics of the separator system engineering design, including loading rate, magnet strength, conveyor speed, material of construction, etc.
- Operational characteristics, including energy requirements, routine and specialised maintenance requirements, simplicity of operation, reliability,
- noise output, and air and water pollution control requirements.
- Locations where ferrous materials are to be recovered from solid wastes.
- Site consideration, including space and height, access, noise and environmental limitations.

Screening

Screening is the most common form of separating solid wastes, depending on their size by the use of one or more screening surfaces. Screening has a number of applications in solid waste resource and energy recovery systems. Screens can be used before or after shredding and after air separation of wastes in various applications dealing with both light and heavy fraction materials. The most commonly used screens are rotary drum screens and various forms of vibrating screens.

Rotary Drum Screen

Note that rotating wire screens with relatively large openings are used for separation of cardboard and paper products, while vibrating screens and rotating drum screens are typically used for the removal of glass and related materials from the shredded solid wastes.

Selection of screening equipment

- The various factors that affect the selection of screens include the following:
- Material specification for screened component.
- Location where screening is to be applied and characteristics of waste material to be screened, including particle size, shape, bulk, density and moisture content.
- Separation and overall efficiency.
- Characteristics screen design, including materials of construction, size of screen openings, total surface screening area, oscillating rate for vibrating screens, speed for rotary drum screens, loading rates and length.
- Operational characteristics, including energy requirements, maintenance requirements, simplicity of operation, reliability, noise output and air and water pollution control requirements.
- Site considerations such as space and height access, noise and related environmental limitations.

$$\text{Recovery (\%)} = \frac{U \times W_u}{F \times W_f} \times 100$$

$$W_f = \frac{\text{Weight of sample}}{\text{Weight of material fed to the screen}}$$

$$W_u = \frac{\text{Weight of sample in underflow}}{\text{Total weight of material in underflow}}$$

The effectiveness of the screening operation can be determined by:

Effectiveness = recovery × rejection

where, rejection = 1 – recovery of undesired material

$$= 1 - \frac{U(1 - W_u)}{F(1 - W_f)}$$

Therefore, the effectiveness of screen is:

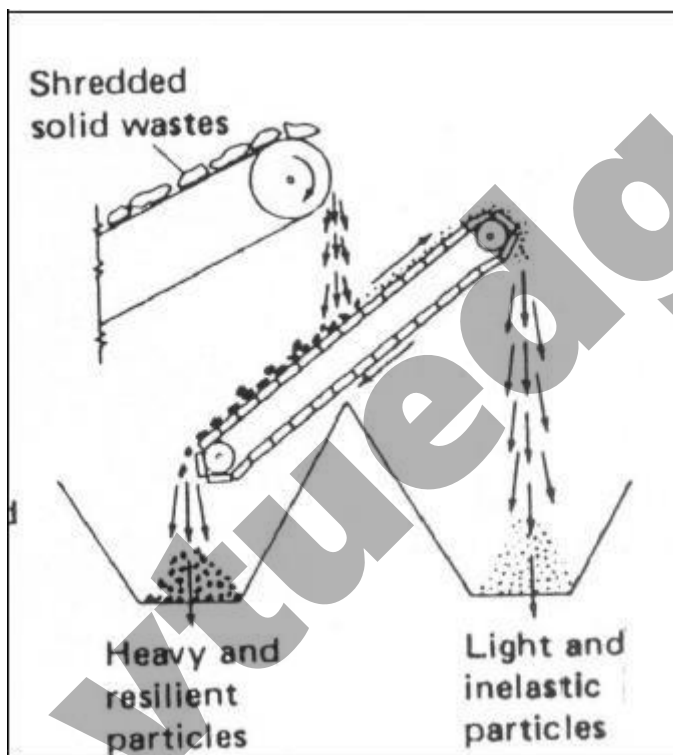
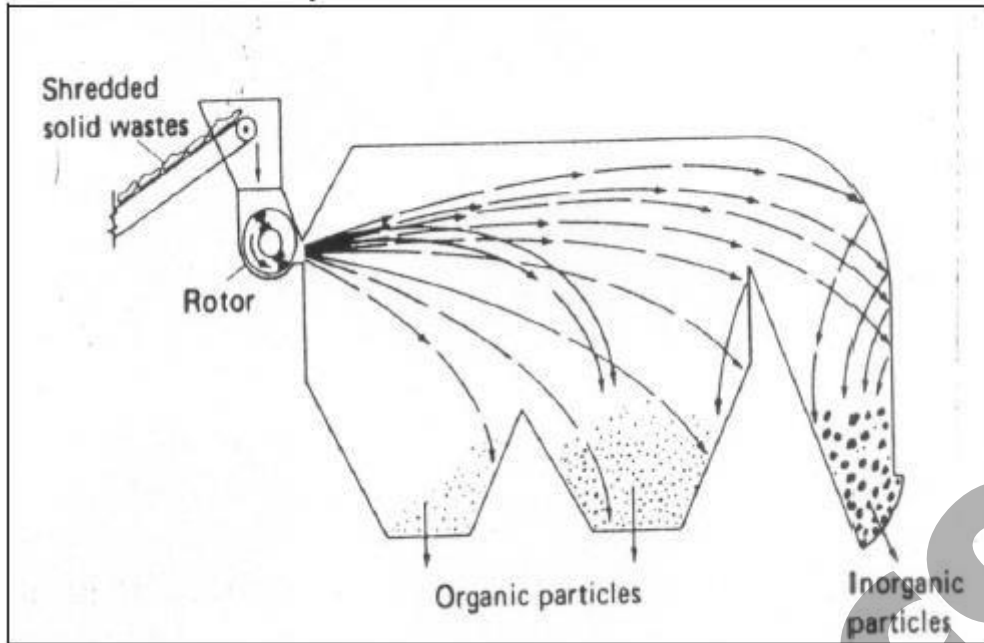
$$\text{Effectiveness} = \frac{U \times W_u}{F \times W_f} \times \left[1 - \frac{U(1 - W_u)}{F(1 - W_f)} \right]$$

Other separation techniques

Besides the mechanical techniques we studied earlier for segregating wastes, there are others. A description of some of these other separation techniques is given below:

(i) **Hand-sorting or previewing:** Previewing of the waste stream and manual removal of large sized materials is necessary, prior to most types of separation or size reduction techniques. This is done to prevent damage or stoppage of equipment such as shredders or screens, due to items such as rugs, pillows, mattresses, large metallic or plastic objects, wood or other construction materials, paint cans, etc.

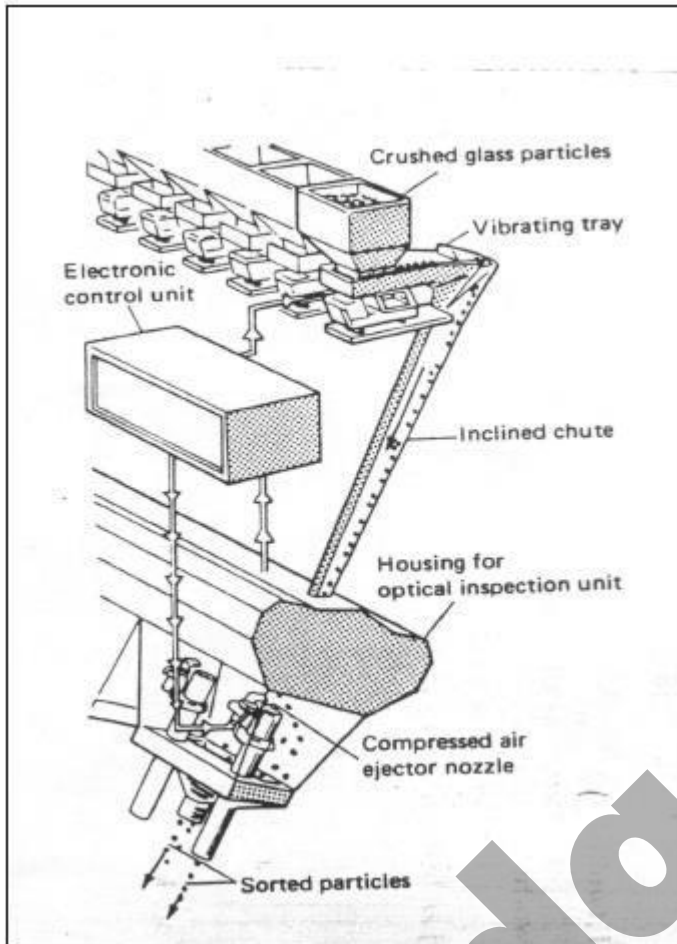
(ii) **Inertial separation:** Inertial methods rely on ballistic or gravity separation principles to separate shredded solid wastes into light (i.e., organic) and heavy (i.e., inorganic) particles.

Ballistic Inertial Separator

Flotation: In the flotation process, glass-rich feedstock, which is produced by screening the heavy fraction of the air-classified wastes after ferrous metal separation, is immersed in water in a soluble tank. Glass chips, rocks, bricks, bones and dense plastic materials that sink to the bottom are removed with belt scrappers for further processing. Light organic and other materials that float are skimmed from the surface. These materials are taken to landfill sites or to incinerators for energy recovery. Chemical adhesives (flocculants) are also used to improve the capture of light organic and fine inorganic materials.

Optical sorting: Optical sorting is used mostly to separate glass from the waste stream, and this can be accomplished by identification of the transparent properties of glass to sort it from opaque materials (e.g., stones, ceramics, bottle caps, corks, etc.) in the waste stream. Optical sorting involves a

compressed air blast that removes or separates the glasses – plain or coloured. An optical sorting machinery is, however, complex and expensive.



So far, we discussed component separation through air classifiers, magnetic separators, screens, and hand sorting, flotation, optical sorting and inertial separators. In case, however, the waste consists of moisture, we need to remove it for efficient management. It is in this regard that drying and dewatering are considered the most appropriate means of removal of moisture.

DRYING AND DEWATERING

Drying and dewatering operations are used primarily for incineration systems, with or without energy recovery systems. These are also used for drying of sludges in wastewater treatment plants, prior to their incineration or transport to land disposal. The purpose of drying and dewatering operation is to remove moisture from wastes and thereby make it a better fuel. Sometimes, the light fraction is *pelletised* after drying to make the fuel easier to transport and store, prior to use in an incinerator or energy recovery facility.

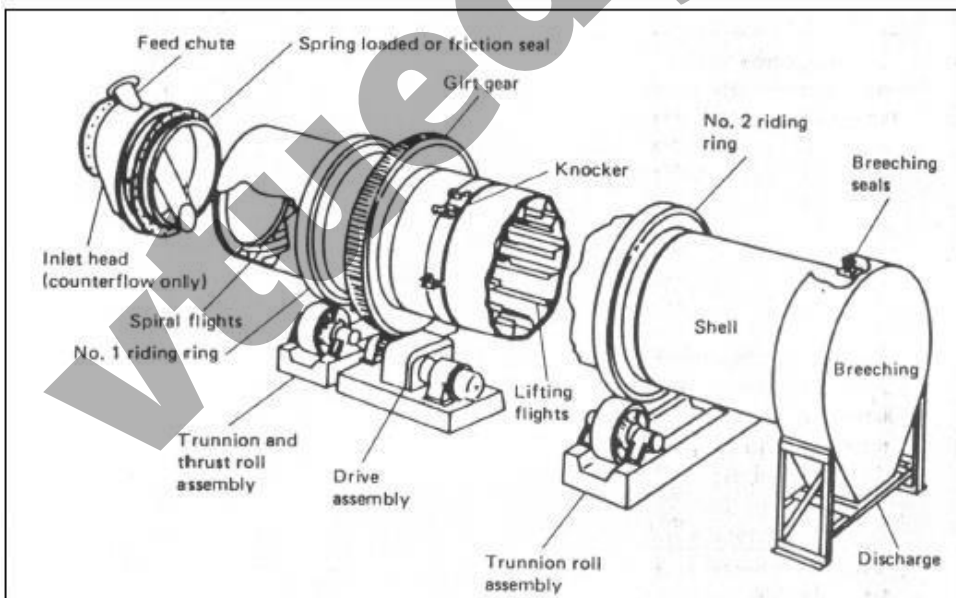
Moisture Content of Municipal Solid Waste Components

Component	Moisture (in percent)	
	Range	Typical
Food wastes	50 – 80	70
Paper	4 – 10	6
Cardboard	4 – 8	5
Plastics	1 – 4	2
Textiles	6 – 15	10
Rubber	1 – 4	2
Leather	8 – 12	10
Garden trimmings	30 – 80	60
Wood	15 – 40	20
Glass	1 – 4	2
Tin cans	2 – 4	3
Nonferrous metals	2 – 4	2
Ferrous metals	2 – 6	3
Dirt, ashes, brick, etc.	6 – 12	8
Municipal solid wastes	15 – 40	20

Drying

The following three methods are used to apply the heat required for drying the wastes:

- (i) **Convection drying:** In this method, hot air is in direct contact with the wet solid waste stream.
- (ii) **Conduction drying:** In this method, the wet solid waste stream is in contact with a heated surface.
- (iii) **Radiation drying:** In this method, heat is transmitted directly to the wet solid waste stream by radiation from the heated body.



Illustrates, a rotary drum dryer is composed of a rotating cylinder, slightly inclined from the horizontal through which the material to be dried and the drying gas are passed simultaneously. The drying of material in a direct rotary dryer occurs in the following stages:

- Heating the wet material and its moisture content to the constant-rate drying temperature.
- Drying the material substantially at this temperature.

- Heating of material to its discharge temperature and evaporation of moisture remaining at the end of the stage.

The retention time in the rotary drum is about 30 – 45 minutes. The required energy input will depend on the moisture content, and the required energy input can be estimated by using a value of about 715 KJ/kg (or 1850 Btu/lb) of water evaporated. Some of the factors, we need to consider in the selection of a drying equipment that include the following:

- Properties of material to be dried.
- Drying characteristics of the materials, including moisture content, maximum material temperature and anticipated drying time.
- Specification of final product, including moisture content.
- Nature of operation, whether continuous or intermittent.
- Operational characteristics, including energy requirements, maintenance requirements, simplicity of operation, reliability, noise output and air and water pollution control requirements.
- Site considerations such as space and height access, noise and
- environmental limitations.

Dewatering

Dewatering is more applicable to the problem of sludge disposal from wastewater treatment of plants, but may also be applicable in some cases to municipal/industrial waste problems. When drying beds, lagoons or spreading on land are not feasible, other mechanical means of dewatering are used. The emphasis in the dewatering operation is often on reducing the liquid volume. Once dewatered, the sludge can be mixed with other solid waste, and the resulting mixture can be:

- incinerated to reduce volume;
- used for the production of recoverable by-products;
- used for production of compost;
- buried in a landfill.

Centrifugation and filtration are the two common methods for the dewatering of sludge. Sludges with solid content of a few percent can be thickened to about 10 – 15% in centrifugation and about 20 – 30% in pressure filtration or vacuum filtration.

SOURCE REDUCTION: BASICS

Purpose:

(i) **Product reuse:** Using reusable products, instead of their disposal equivalents, reduce the amount of materials that are to be managed as wastes. An example of product reuse is the reusable shopping bag.

(ii) **Material volume reduction:** Reducing the volume of material used changes the amount of waste entering the waste stream. This helps in controlling the waste generated and its disposal. For example, buying in bulk or using large food containers reduces the amount of packaging waste generated.

(iii) **Toxicity reduction:** Source reduction reduces the amount of toxic constituents in products entering the waste stream and reduces the adverse environmental impacts of recycling or other waste management activities. For example, substitution of lead and cadmium in inks (solvent-based to waterbased) and paints is a source reduction activity.

(iv) **Increased product lifetime:** Source reduction facilitates the use of products with longer lifetime over short-lived alternatives that are designed to be discarded at the end of their useful lives. Put differently, it encourages a product design that allows for repair and continued use rather than disposal. Manufacturing long-life tyres is a good example of increasing product lifetime.

(v) **Decreased consumption:** This refers to the reduced consumption of materials that are not reusable (e.g., using a reusable shopping bag instead of picking up plastic bags from the store). Consumer education about the materials that are difficult to dispose of or are harmful to the

environment is essential. Buying practices can thus be altered (e.g., buying in bulk) to reflect environmental consciousness. In brief four main advantages of source reduction are

- Reduction in extent of environmental impacts
- Reduction in resource consumption and generation of pollution
- It includes producer, consumer, prudent and efficient activities.

Implementation:

There are several specific actions that can take place at the local level to encourage source reduction, some of which are given below:

(i) **Education and research:** Consumers, businesses, industries, schools, etc., can implement education and research activities to address the need for source reduction, its consequences, available choices, benefits and costs. Essentially, the aim of such education and research activities is to provide and develop information about source reduction goals, needs and methods and to elicit voluntary efforts from the public and private sectors to help bring about some specific changes. Some of the activities that reflect education and research to encourage source reduction include:

- forming stakeholder councils (industry, government agencies, etc.) to develop a source reduction message to the public and to carry out educational and research activities;
- exploring and developing funding sources, such as government grants, financial support for industries, private funding grant, direct tax and solid waste surcharge;
- developing media campaign for public outreach, including posters, conferences and forums concerning source reduction;
- developing curricula for schools and universities as well as organising a group of professionals with knowledge of source reduction and solid waste management.

(ii) Financial incentives and disincentives:

Linking an economic benefit to the implementation of source reduction activities encourages source reduction. For example, financial disincentives represent additional costs to the waste producing activities that could be avoided through source reduction activities. The various measures that can be targeted at consumers and industries include the following:

- Tax credit or exemption may be given to companies and institutions that follow specific source reduction procedures for manufacturing and consuming.
- As variable waste disposal charges for garbage collection are generally taken for post-consumer solid waste, the fees or payment can be based on the number of garbage cans used, number of bags collected or the frequency of collection.
- Graduated fee structure for garbage collection can be based on quantity and frequency.
- Product disposal charges can be assessed either on the producers at the time of manufacture or on the consumers at the time of purchase. However, we need to note that though these charges can encourage source reduction on economic grounds and the funds generated from the charges can be used to correct and reduce the impacts of product disposal, it is difficult to assess such charges effectively and efficiently.

(iii) **Regulation:** Although most regulation occurs at the national and state level, local authorities can participate in legislative activities in developing regulations that affect municipal SWM. It is possible, for example, to establish a programme to inform the consumers about environmental impacts,

durability, reusability and recyclability of products as well as to declare source reduction as a top priority in SWM. Regulatory option of source reduction can include the following:

- Quantity control regulation, which encourages substitution of products that have the same function but pose less threat to human health and environment, through restrictions and bans.
- Product design regulation, which includes products that do not meet certain design criteria and could be subjected to quality control by sales tax and restrictions.

Monitoring:

Monitoring facilitates the evaluation (i.e., efficacy and efficiency) of source reduction, the identification of possible source reduction measures and programme revisions and the obtaining of funds and resources for source reduction initiatives/programmes. Monitoring should, therefore, be an integral part of a source reduction programme. We must, however, note that source reduction is more difficult to measure on a broad scale than other methods of SWM. When several waste reduction techniques are used simultaneously, for example, it is not easy to determine which portion of the diversion was due to source reduction.

Source reduction often results in substantial and measurable cost savings from activities such as waste collection, transportation and disposal, and direct savings. In addition, source reduction is cost-effective in decreasing pollution, purchase, use, and regulatory compliance costs. It also reduces product and material use and disposal costs in the manufacturing process, making business operations more efficient. There is, however, some concern that source reduction might reduce economic growth by decreasing consumption. Costs associated with source reduction must, therefore, form part of the monitoring process to ensure Accountability.

Life cycle analysis

Ideally, to assess and quantify these trade-offs, a life cycle analysis (LCA) – variously known as life cycle assessment, ‘cradle to grave analysis’ or ‘econ-balance’ – is required, which involves the evaluation of some aspects, particularly most often the environmental aspect, of a product through all stages of its life cycle.

LCA, in essence, takes a detailed look at all resources used and the products and by-products generated throughout the entire life of a product or process. The analysis begins with raw materials and energy acquisition, examines manufacturing and product fabrication, filling, packaging and distribution, and consumer use and reuse, and finally ends with an analysis of waste management.

Note that LCA represents a rapidly emerging family of tools and techniques designed to help in environmental management and long-term sustainable development (European Environment Agency, 1998) and the LCA procedures are being developed to assess the overall environmental impact of products and their packages.

Evaluation

Before adopting source reduction policies, it is important that we develop a framework for evaluating various options. Some of the criteria to be considered in this regard are:

- Social and economic equity.
- Economic and administrative feasibility, efficiency and cost.
- Volume requirement and scarcity of materials and natural resources in
- product manufacture.

- Volume of product and its by-products that must be eventually disposed.
- Useful life, reusability and/or recyclability of a product.
- Priority of source reduction of more hazardous products to less hazardous ones.

SIGNIFICANCE OF RECYCLING

Recycling is perhaps the most widely recognised form of source reduction involving the process of separating, collecting, processing, marketing and ultimately using a material that would have otherwise been discarded.

This form of source reduction, i.e., recycling, is similar to other forms, in that it: ultimately using a material that would have otherwise been discarded. This form of source reduction, i.e., recycling, is similar to other forms, in that it:

Recycling is one of the fundamental parts of the waste management plan. Although it alone cannot solve a community's municipal SWM problem, it can divert a significant portion of waste stream from disposal in landfill and combustion facilities. Recycling has a lot of direct and indirect significance for the society, and this can be grouped under the following three broad areas:

(i) Economic significance:

Economic assessment of waste recycling is a difficult task as many of the beneficial environmental and social impacts of recycling are long-term and are intangible, and, therefore, are difficult to quantify. Some of the short- and long-term economic benefits are:

- **Cost reduction:** Resource recovery through recycling of solid waste could be of interest to waste management authorities as a means of reducing the waste disposal cost. Any saving in waste management cost could be a significant incentive to the authorities to increase the coverage of service areas and improve the service level. They can save cost from fuel for transportation, operation and maintenance, and generate revenue by sale of recyclables, etc.
- **Employment:** Recycling of waste is a labour intensive activity, and its potential to ease the unemployment problem is high. Enhanced recycling activities, for example, can create an additional job market for skilled and unskilled workforce, and they can adapt to any of the occupations such as a labourer in recycling business or industry or a dealership.
- **Energy saving:** Use of recyclables in some industrial processes is known to consume less energy than the use of any other raw material. The reduction in energy consumption in one industry could mean its availability for some other industry in need.
- **Reduced health care costs:** Improved health and sanitary conditions in urban areas resulting from indirect benefits of waste recycling can reduce the investment in public health programme.
- **Saving costs for other public utilities:** Enhanced solid waste recycling practices can reduce the frequency of sewer clogging, blocking of natural watercourses and pollution of water bodies. This will benefit the concerned public utilities through reduced cost in cleaning sewers and improved public safety due to blocked sewers and narrowing of natural watercourses.

(ii) Environmental and health significance:

The volume of waste is increasing rapidly because of population growth and economic development. The composition of waste is also changing, leading to waste production with more recyclables. At the same time, polluted waste fractions are increasing because of increasing complex processes being used in industries, and these contribute increasingly to environmental degradation. This notwithstanding, recycling helps, among others, in the following ways, to facilitate effective waste management:

- **Improved environment:** The environmental pollution may be due to inadequate SWM as well as due to its effect on other urban infrastructure. Recycling reduces the volume of waste that has to be finally dumped, and thereby causing reduction in pollution at the waste disposal sites. When there is reduction in volume of waste because of its increased reuse, different types of pollution (e.g., water, air and land) will get abated.
- **Natural resource conservation:** Industries with natural products as their raw material for production are depleting natural resources. Use of more and more recyclable solid wastes in industrial production will relieve the tremendous pressure on these precious resources. For example, recycling of waste paper means a lower demand for wood, which means less cutting of trees and an enhanced possibility for sustainable use of the forest. Using recyclable items in the production process would reduce the demand for energy as well.

(iii) Social significance:

People engaged in waste collection activities are normally of low social and economic standing. This is especially true with scavengers, which is evident from persisting poor quality of their living and working conditions. Different groups of people engaged in waste recycling have a hierarchical social and economic status, in which, processors are at the top of the hierarchy followed by waste dealers and wholesalers, waste buyers and waste collectors in that very order, while scavengers are at the bottom. Although there is this social and economic hierarchy within the waste recycling business, the overall social esteem of waste recycling operators is low.

A formal recycling arrangement will help promote the social esteem of waste workers and facilitate their upward social mobility due to increased earning. In addition, the improved recycling activity will increase the economic value of the waste and will reduce waste scavenging activity providing opportunity for scavengers to switch to a more socially acceptable occupation. In short, institutionalised recycling programmes will help remove the stigma associated with waste scavenging and transform it to an economic enterprise.

PLANNING OF A RECYCLING PROGRAMME:

Numerous recycling options are available, and recycling programme development requires strategic planning. Planning for recycling involves understanding markets, assessing local expertise, setting goals and fostering public participation. An efficient recycling programme requires a systematic approach to all programme components, which are interrelated, and therefore, decisions about one must be made taking into consideration other components. As a successful recycling requires public participation, programmes must be designed keeping in view public convenience and support.

(i) Build local expertise: Small projects help build local expertise in recycling and minimise the problems associated with poor planning. With smallscale projects, it is easy to compare and evaluate the programmes and techniques that are considered most successful within the community. When the time comes to develop a large-scale programme, there will be practical

experience and an established decision-making framework, which will enhance the programme's success.

(ii) **Understand and develop a recycling market:** While planning for a recycling programme, it is important to find an outlet for the recyclable material. Market analysis is both a planning and ongoing activity, as even the most successful recycling programme can be severely affected by market fluctuations. Recycling programmes must, therefore, be designed with the flexibility to handle fluctuating markets and uncertain outlets for material.

(iii) **Foster public education and involvement:** Public participation is one of the most important factors deciding a programme's success. The public has a right and a responsibility to understand the full costs and liabilities of managing the waste they produce. A well-planned public education and involvement programme will foster public interest in recycling.

(iv) **Assess local waste stream:** Planning any recycling programme requires the knowledge of the local waste stream. Choosing the right material to recycle and designing the logistics of the programme are the important parts of the planning process.

(v) **Augment existing programme:** Recycling should augment the success that has been attained by other groups operating recycling programmes. This is very important for planning and success. Other programmes may be run by local volunteer organisations to raise funds or as a community service.

(vi) **Set goals and objectives:** Part of the planning process involves setting goals and objectives. The preliminary assessment of waste stream helps in deciding long-term goals for a community. Planning objectives may include determining the type of waste stream component that should be programmed, investigating the feasibility of the curbside (kerbside) programme, public outreach avenues, etc. The community will benefit from carefully developed achievable goals and objectives, and from an integrated approach to waste management.

(vii) **Coordinate the programme:** Recycling programme is considered a public service. Therefore, local governments are required to ensure that all services are provided properly. Like any other public service, recycling programmes should be consistent, predictable, equitable and efficient.

(viii) **Evaluate the programme:** New programmes and technologies are evolving continuously, which make the planning for recycling an ongoing process. This requires experiment and evaluation. Even the best recycling programmes experiment with new techniques to improve on their current efforts. If recycling programmes are properly planned and implemented, they would then add to the overall municipal waste management activity.

RECYCLING PROGRAMME ELEMENTS

Recycling programmes are designed according to the needs and priorities of the communities. Elements of a recycling programme include source separation, curbside (kerbside) collection, material resource facilities and full stream processing. Recycling, generally, has a positive impact on other municipal waste management programmes. This may include a mix of strategies, ranging from simple, single material drop-off centres to large scale, centralised processing facilities.

(i) Source separation

Source separation refers to the segregation of the recyclable and reusable materials at the point of generation. This requires that several designated materials be separated into their own specific

containers, while other programmes use only two containers – one for the storage of mixed recyclables and the other for regular wastes. Source separation may be voluntary or mandated and is done in conjunction with several recycling programmes.

(ii) Drop-off/buy-back

A drop-off programme requires residents to separate the recyclable materials and bring them to a specified drop-off or collection centre. However, we must take into account local conditions when designing a collection programme. For a community that does not provide a curbside pickup, for example, educating and encouraging citizens to deliver materials to a drop-off site may be all that is needed. A recycling centre can be established at the same location where residents deliver waste. Mobile recycling drop-off trailers can also be used. Dropoff recycling, however, is less convenient than curbside pickup. If a thorough educational and promotional effort is not made, drop-off programme tends to have lower participation rates than curbside collection.

Buy-back refers to a drop-off programme that provides monetary incentives to participate. In this type of programme, the residents are paid back for their recyclable material directly or indirectly through the reduction in collection and disposal fees. Establishing a buy-back centre (i.e., a place where recyclables are purchased) may help induce citizens to recycle. Some buy-back centres purchase some materials and accept others, depending on current market conditions. Private or public mobile buy-back operations can serve some areas, purchasing recyclables from small communities or from neighbourhoods of large metropolitan areas on a regular schedule.

(iii) Curbside programme

In a curbside system, source separated recyclables are collected separately from regular refuse from the curbside, alley, or commercial facility. Curbside programmes vary greatly from community to community. Some programmes require residents to separate different materials that are stored in their own containers and collected separately. Other programmes use only one container to store recyclables or two containers, one for paper and the other for heavy recyclables (e.g., glass aluminium, etc.).

Collection vehicles for recycling

Collection vehicles that are designed specifically for collecting recyclables have several storage bins, which can be easily loaded and often equipped with automatic container-tipping devices. Although these modified vehicles may still be considered as options, a dedicated, closed-body collection vehicle for recycling with sufficient capacity offers such significant advantages as easy loading and unloading, flexible compartments and protection from weather. Of course, this warrants a substantial initial investment.

Processing equipment for recycling

Recycling involves a number of processing techniques and these processes require different equipments, some of the special equipments used in recycling are:

- **Balers:** Balers can be used to densify many types of materials including paper, cardboard, plastics and cans. Balers can improve space utilisation and reduce material transportation costs.
- **Can densifiers:** Can crushers are used to densify aluminium and steel cans prior to transport.
- **Glass crushers:** These are used to process glass fraction separated by colour and break it into small pieces. This crushed material is then called cullet, and can be reprocessed into new glass products.

- **Magnetic separators:** These are used to remove ferrous material from a mixture of materials.
- **Wood grinders:** These are chippers and are used to shred large pieces of wood into chips that can be used as mulch or as fuel.
- **Scales:** These are used to measure the quantity of materials recovered or sold.

Material recovery facilities (MRF)

MRF (pronounced 'murf') is a centralised facility that receives, separates, processes and markets recyclable material. It can be operated with both drop off and curbside programmes. The primary advantage of MRF is that it allows materials directly from the municipalities and processes them uniformly. It is generally designed to handle all type of recyclables. Implementation of MRF in a municipality depends upon a number of factors as follows:

- **Market demand:** When additional processing is required, MRF is more useful as buyers may have certain material specifications.
- **Separate collection:** In systems that require residents to separate their recyclables, intermediate separation and processing is required.
- **Number of different recyclables:** In general, a MRF will be more beneficial when a large number of different recyclables are collected.
- **Quantities of materials:** Because MRF involves substantial capital and operating costs (e.g., buildings, equipment and labour), it is expected to handle a significant amount of materials to justify its operation.

Full stream processing

This is a high technology separation technique, which processes all components of municipal waste. The materials recovered by this process tend to be of lower quality than those recovered or source separated in MRF because the former is a mix of various types of wastes. To achieve a better quality, the materials obtained through the full stream processing must be cleaned, which is a costly process. However, this technique remains attractive because it does not require source separation, and it is used in the following applications:

- (i) Refuse derived fuel (RDF) preparation: In this application, it is used to extract the combustible portion of municipal waste.
- (ii) Municipal waste composting: In this application, it is used to concentrate the compostable portion of municipal solid waste. Note that this is sometimes performed as part of RDF preparation.
- (iii) Material recovery: In this application, it is used to recover and resell certain materials, and thereby making material recovery a recycling technology as well. In full stream processing, depending on the facility design, the materials are separated either mechanically or by hand, and size and weight are the main characteristics used to separate the materials.

For example:

- when the material is dumped, oversized materials such as furniture, etc., are removed;
- rotating screens are used to separate materials of different sizes (small and large);
- ferrous material is extracted using a magnet system;
- air classifier is used to separate the lighter material;
- light materials including plastic and paper are further processed into RDF;
- heavy fraction is mechanically or manually sorted to recover saleable materials such as cardboard, etc.

COMMONLY RECYCLED MATERIALS AND PROCESSES

(i) Paper and cardboard

Paper and cardboard form the second biggest component of domestic waste after organic waste, and contribute to about 13% of the total domestic solid waste (UNCHS, 1994). Paper recycling is one of the most profitable activities and is practised extensively. It reduces the demand for wood and energy and helps solve littering problem in the city and around dumping site. It has an acceptable working condition and health risks are limited. Recovered paper and paper products are bought and sold through a well-established network of local processors and vendors who typically bale these materials for sale. Of late, paper mills have started buying directly from the collectors.

The paper industry is making a significant investment in manufacturing capacity for making paper and paper products with recycled content. Recovered paper is classified as newsprint, corrugated cardboard, mixed paper (including magazines, junk mail and cardboard), high-grade paper (white office paper, photocopying paper), and pulp substitute paper (usually mill scrap). Paper mills, the most common end users of recovered paper, use the material as a feedstock to manufacture recycled paper and paper products, such as newsprint, chipboard, craft linerboard, corrugating medium, roofing felt and tissue products. Shredded paper is used to make animal bedding, hydro mulch, moulded pulp products and cellulose insulation.

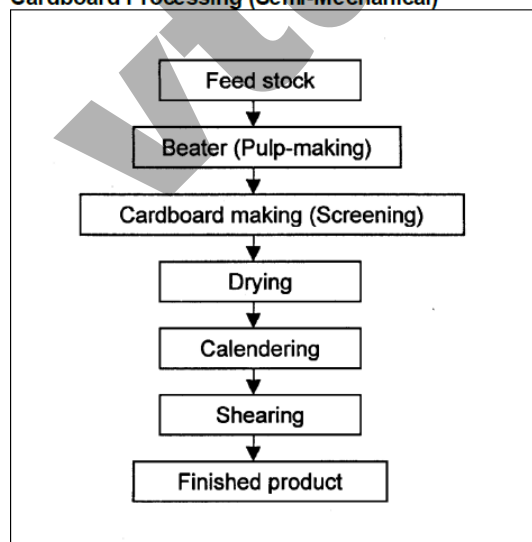
A typical input-output model of the paper processing technique consists of the following:

- Material inputs: Paper/cardboard scrap, magazines, newspapers, computer paper, wrapping paper, craft sacks, cartons, etc.
- Product outputs: Paper sheets, boxes, filter paper, mosquito mats (to absorb chemical repellent), merchandising packets, decorative items, etc.

And, the technology in the recycling of paper and cardboard involves the following processes:

(i) Cardboard processing (semi-mechanical) plant: In semi-mechanical plants, paper scrap is pulped in a beater machine. The paper pulp is spread on a rotating sieve and pressed mechanically. Cutting is done manually and after cutting and sun-drying, the cardboard is calendered and sheared into sheets, from which boxes for shoes, sweets, etc., are made:

Figure 10.1 Cardboard Processing (Semi-Mechanical)

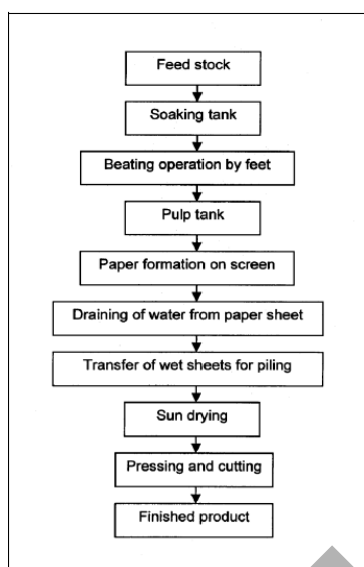


The various machineries used in the processes include:

- Beater: This consists of a masonry and concrete tank and a mild steel beater roll, driven by an electric motor.
- Sieving screen: This separates materials of different size.
- Cardboard making unit: This is a nylon conveyor screen driven by an electric motor.
- Calendering machine: This is used to finish the surface of paper and is electrically operated.

(ii) Hand-made paper: In hand-operated units, pulping and beating of paper is done manually in an open masonry or concrete tank. The material is sieved in a second tank and diluted with water to a specific consistency. A wooden framed screen is dipped in the tank in order to form an even layer of the wet pulp to a desired thickness (over the screen mat). The layer is skillfully removed and transferred from the mat and the sheets are sun dried on smooth walls or on other smooth surfaces. The dried sheets are then pressed and cut into required sizes. The calendering is performed in an electrically operated machine.

Hand-Made Paper Processing



(iii) Glass

Glass is one of the most commonly recycled materials, and the market for post consumer glass has historically been steady. Glass generally accounts for 2.5% by weight of the total solid waste generated (UNCHS, 1994). Though it does not contribute to the environmental problem, glass does cause a serious problem of littering. The economic impacts are cost of waste collection and disposal, reduction in use of natural products and energy consumption. Recycling of broken glass reduces the risk of diseases caused by cuts and wounds. Glass recycling is a labour intensive process and provides employment opportunity. Glass is typically broken for size reduction or crushed and ultimately sold to glass manufacturers as furnace-ready cullet after metal caps, rings, labels, etc., are removed.

Glass manufacturers purchase glass for reprocessing into new, clear, green and brown glass jars and bottles. The market for recovered glass has been strong and stable for brown and clear containers. Green glass, however, is seldom used to package goods domestically, so fewer companies produce glass of this colour. Alternative markets for glass include art glass, sandblasting, and industrial windowpane glass and fibreglass insulation. Although the glass industry has made a commitment to increase the demand for recovered glass, there is an

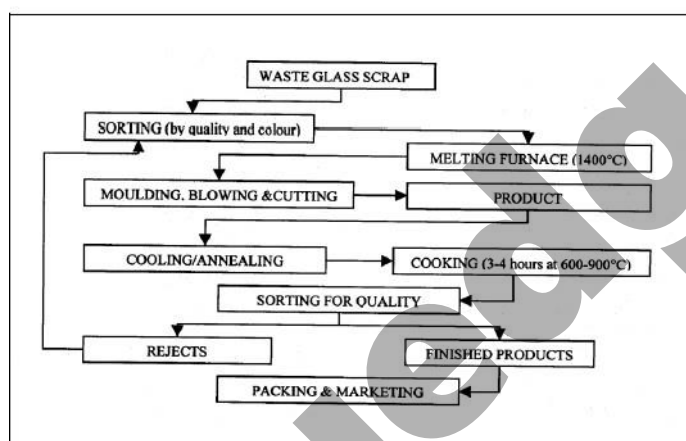
important and pervasive market concern about the quality of material being produced by collection programmes and processing facilities.

Recycling programme planners must address this concern for high-quality recovered glass as well as for other commodities. Typically, while glass scrap is the material input, glass products are the product outputs in the glass recycling technique which involves a semi-mechanical process. This, we will study next.

Semi-mechanical process

The waste glass cullet is sorted according to colour and melted in an oven at 1400 C. The oven used in a small production unit is locally made from fire clay and known as pot furnace. To improve the unbreakability of the glass product, chemicals such as soda ash, potassium carbonate, borax, lime, etc., are added to the cullet before melting. When the metal has completely melted, the temperature is raised to refine the glass. After refining, the temperature is lowered to carry out the moulding operation. After the glass takes the shape of the mould and hardens, unwanted portions are cut and removed, and sharp edges are smoothened in a machine. The product is then annealed by cooling and heating for three to four hours at 600 – 900 C to reduce the brittleness of the glass. After quality inspection, defective glass products are returned to the recycling process.

Glass Processing (Semi-Mechanical)



The following machinery is used in the process:

- Furnace: It is used to melt the glass cullet. The glass cullet is melted at 1400 C (locally fabricated) and the product obtained is annealed at 600 C (locally fabricated).
- Semi-mechanical die: It is used to cast the die. It uses both mechanically and manually operated moulds.
- Air compressor: It is used for blowing molten glass into dies.
- Printing machine: It is used for printing trademark.

(iv)Metals

Ferrous metals like iron, steel, etc., and non-ferrous metals like aluminium, copper, zinc, lead, silver, etc., are some of the metals, which exist in the waste stream. On an average, metals account for 2% of total solid waste generated (UNCHS, 1994). Extraction of metals from natural ores depletes the mineral resources. Metals when dumped at landfill sites produce hazardous leachate with heavy metals in solution.

Using recycled metals substantially reduces operating costs of industries. Metal scrap is cheap and the energy consumption is lower when products are manufactured from scrap. The long-standing track record makes ferrous and non-ferrous metal market among the most stable of all recyclable materials. Ferrous and non-ferrous metals can be bought and sold through processors and vendors. Ferrous scrap includes household appliances, equipments, cans, and other iron and steel products. Non-ferrous scrap metals include aluminium, copper, lead, tin, etc. Both ferrous and non-ferrous metals can be prepared for sale through some combination of processing by flattening, baling, and shredding of the material. In some cases, processors melt the metal into ingots before selling it to end-use markets. Several foundries and steel mills have begun or expanded recycling efforts, and steel mini-mills also appear to be increasing their use of recovered steel in regions, which typically lack large mills.

The typical material inputs and product outputs in this industry are the following:

- (i) Material inputs: Aluminium, brass, copper, zinc, tin, iron, steel, etc.
- (ii) Product outputs: Sanitary and gas fittings, funnels, buckets and storage bins, reinforced steel bars, hand tools, etc.

Metal processing

Now, we will discuss the process involved in recovering metals. Most of the recovered metals are processed by big industries. Ferrous metals are processed by iron industries to produce iron bars, channels, angles, etc. Local artisans process part of the ferrous metal in many cities. In a small-scale cottage industry, a particular kind of metal (ferrous or non-ferrous) is melted in a crucible in the coal furnace and the molten metal is cast into the desired mould to make ingots of required shapes and size. New and melted recycled metals are mixed together in a 3:1 ratio for better quality products. Ingots are sold to manufacturers to obtain different products from the metal. Local artisans heat the iron ingots in a coal furnace and beat them into different shapes for various kinds of tools and implements. Annealing hardens the cutting edges of the tools. The tools used in the process include coal burner, furnace (coal fired), moulding gadgets, dies (pattern) and auxiliary tools.

(v)Plastic

These days, plastic is posing serious littering problem in cities and around collection points and dumping sites. With an average 8% by weight of the total amount of domestic waste, plastic is one of the major constituents in waste stream (UNCHS, 1994). Un-recycled plastic, when burned, contributes to greenhouse gases. The direct benefits of recycling plastic waste are reduction in the cost of raw material and energy saving. Plastic recycling also helps in employment generation along with reduction of volume transport and space requirements for dumping.

Most plastics are densified locally by flattening, baling, or granulating, and sold either to converters, where the resins are turned into pellets, or directly to domestic or export end users for remanufacture into products such as bottles, carpet and carpet backing, flower pots, and insulation material.

Post consumer plastic-resin recycling technology has developed more rapidly than technologies for any other recovered material in the last half century. Only five to ten years ago post consumer high-density polyethylene (HDPE) and polyethylene terephthalate (PET) plastics were vaguely considered recyclable.

These two resins, especially HDPE milk jugs and bottles and clear PET plastics, now hold a stronger place in the market. End uses for recycled HDPE include non-food bottles, drums, toys,

pipes, sheets and plastic pallet, and for PET include plastic fibres, injection moulding, non-food grade containers and chemicals. The recyclability of other resins, such as polystyrene, polyvinyl chloride, low-density polyethylene (LDPE), polypropylene and mixed plastic resins is making strides but much remains to be done. The input materials, the output products involved in the recycling of plastics are given below:

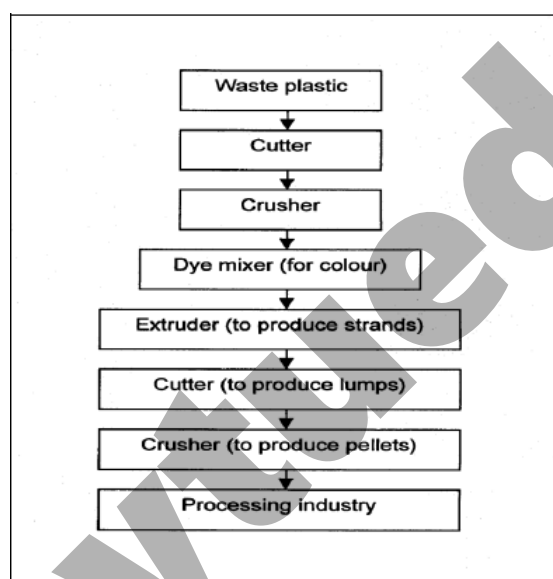
- (i) Material input: Plastic scrap (thermoplastic and thermosetting).
- (ii) Product output: Toys, boxes, slippers, shoes, pellets, buckets, cans, etc.

We will now discuss the processes involved in plastic recycling.

Plastic processing

In plastic processing, the primary steps are sorting by colour and quality, and cutting and crushing the sorted material. The crushed product (granules) of plastic is melted, colour dyed and manually moulded into a cheaper product. The poly vinyl chloride plastic is blended with a specific colour dye in a mixing machine. The coloured material passes through an extruder machine to produce thick plastic strands. The strands are manually cut into lumps and these are used for manufacturing items either manually or mechanically.

Note that in the manual process, the lumps are further chopped into smaller pieces and melted. The melted material is moulded into products such as shoe soles, toys and boxes. Machineries used in this process are electrically operated crusher, extruder, mixer and manually operated moulding machine (in which material is electrically heated at 359 C).



(vi) Batteries and tyres

Battery recycling is not only a response to market condition (i.e., price of lead) but also is important due to concern over the toxic compound including lead, cadmium and mercury present in many batteries. Like other materials, battery recycling depends largely on market conditions and requires consistent collection and processing. Household batteries come in a variety of types including alkaline, carbon, zinc, silver, nickel, cadmium, etc. Only those containing mercury and silver are marketed to end users, who extract metals. Automobiles use lead acid battery, which contains lead and sulphuric acid, both hazardous materials. Battery reprocessing includes breaking open the batteries, neutralising the acid, chipping the container for recycling and smelting the lead to produce recyclable lead.

Tyres represent a special challenge to solid waste and recycling programme managers. The use of chipped or shredded tyres as a source for fuel is growing. Electricity-generating facilities, pulp and paper mills and cement kilns are the most common processes using scrap tyres (EPA, 1989 and 1995).

CASE STUDY: SOURCE REDUCTION AND RECYCLING IN BANGALORE

Source reduction, including reuse and recycling, can help reduce waste disposal and handling costs, because it avoids the costs of municipal composting, landfilling and combustion. Source reduction also conserves resources and reduces pollution, including greenhouse gases that contribute to global warming. Waste reduction, reuse and recycling, thus, play an important role in SWM. In what follows, we present the statistics on waste recovery and recycling done in Bangalore, India.

In Bangalore, 66% of the waste generated is collected for recovery, i.e., about 2,373 tonnes per day. While 722 tonnes per day is reused, the rest (i.e., 1,450 tonnes) goes for recycling. The agents involved in the collection and recovery of wastes in the city include waste pickers, IWB (i.e., itinerant waste buyer), middlemen (or intermediaries), the municipality and recycling units (both large and small). While the three agents in the informal sector and the municipality are directly involved in waste collection activities, the waste is processed by the recycling units, which receive recyclable waste from middlemen and municipality.

Waste Recovery and Collection by Stakeholders

Stakeholders	For reuse	For recycling		
		Waste pickers	IWB	Intermediaries
Households	101.7	74.5	60.4	57.7
Commercial establishments	350.8	227	0.0	24.3
• markets	112.7	200	0.0	8.8
• hotels	238.1	27	0.0	15.4
Institutes	0.0	9.6	0.0	61.3
• hospitals	0.0	1.0	0.0	0.0
• offices	0.0	8.4	0.0	0.0
• educational institutions	0.0	0.2	0.0	61.3
Industries	269.3	0.8	0.0	934.5
Total	721.8	312	60.4	1077.8

Of the 1450 tonnes collected for recycling, 1077.8 tonnes come from intermediaries, 60.4 come from IWB and 312 tonnes come from waste pickers. This amounts to 40% of the total waste (i.e., 3613 tonnes per day) generated.